Two Routes to Smaller Diophantine Consistency Equations

Weighted transport and reuse of Pell witnesses

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Abstract

We construct an explicit equation in 83 unrestricted integers of exact degree 624, through a direct numerical proof calculus, and an explicit equation in ten strictly positive integers of exact degree 190. Using the named inherited logical calibrations, ZFC proves that each equation's insolubility in its stated domain is equivalent to Con(ZFC). The direct construction reuses the quotient of a large Pell solution as its modular center witness, then applies certified acyclic unit elimination. It uses neither K215 simulation nor Jones's universal polynomial. Its factored evaluator uses 1,789 binary operations and integer literals of absolute value at most 64. The ten-variable construction changes a positive transport weight so that a center for an entire Pell product works in a newer free-quotient chart. It retains the native/K215 coefficient ancestry and uses 118 live binary arithmetic operations. A companion has degree 202 and 99 operations.

The archive supplies literal equations, all fixed coefficients, finite reconstruction, exact algebra checks, and reproducible code. The proofs are ordinary mathematical arguments composed with identified inherited interfaces; no new end-to-end Lean/Rocq formalization is claimed. The results establish neither consistency nor an actual zero, and do not establish a global simplicity minimum.

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1 Results, comparison, and logical scope

Fix the effective syntactic presentation of ZFC used by the repository's refutation calibration. Its consistency sentence says that no finite proof in that presentation derives a contradiction. Every quantified integer below is an integer internal to the ambient theory; no restriction to externally standard proof codes is made.

There are two main outputs, with different domains and costs. The direct route produces the polynomial D_{83} described in Section 8 and fully specified in Appendix C and its literal electronic appendix. It has 83 unrestricted integer inputs and exact degree 624. The second route gives the following theorem.

Theorem 1.1 (Ten positive variables)**.** *Let P_{190} be the polynomial defined in Section 2, with the four fixed integers specified in Appendix A. Then*

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}_{>0}^{10}\ P_{190}(\boldsymbol{x}) = 0\right) \iff \mathrm{Con}(\mathrm{ZFC}).$$

Its exact total degree is 190*. Its supplied literal circuit uses* 73 *multiplications,* 24 *additions, and* 21 *subtractions.*

The proof composes the new arithmetic arguments with the inherited calibration identified in Section 3. The coefficient specification is independent of the truth of consistency. For the direct route the analogous theorem quantifies over $\mathbb{Z}^{83}$; that domain distinction is substantive.

Ten-positive-variable construction	Degree	Operations	Root scale
Source 172, free quotient and inverse correction	214	87	1
Source 178, whole-product center in older chart	208	116	4
Incoming Affine Horizons, centered free chart	206	90	1
This report, quotient-center companion	202	99	2
This report, weighted whole-product center	**190**	**118**	**4**

The comparison snapshot is the immutable repository commit `cdfea7d90176775d74c1c30afbe49fa78ade9897`. The incoming archives at that snapshot were inspected as well as the placed packages. Thus degree 206, rather than the placed degree 208, is the relevant preceding ten-variable degree benchmark [1, 2, 3]. The drop $206 \to 190$ is 16 degrees, approximately 7.77%. The operation count rises $90 \to 118$; neither point dominates the other under both measures.

The word *simplest* requires a comparison rule. The same repository already contains nine-positive-variable equations of much higher degree, ten-unrestricted-integer equations using a different reduction, and quartics with more variables and large fixed coefficients. For example, package 101 supplies a nine-positive degree-997,810 equation, and package 82 a quartic in 37 unrestricted integers. The new results therefore establish measured improvements in identified families, not minimum variable counts or global optimality. For the ten-positive route the coefficient cost is explicit: the largest port has 592,884 bits. Counting multiplication by that integer as one operation is not a bit-complexity assertion.

To obtain a ten-variable equation over nonnegative integers without changing total degree, replace every input x_j by $x_j + 1$. That translated circuit has ten additional additions. The positive-domain theorem cannot simply be relabeled as an unrestricted-integer theorem.

2 The ten-positive-variable polynomial

The independent inputs and their entire domain are

$$(J, F, \alpha, i, T, w, s, n, \eta, \zeta) \in \mathbb{Z}_{>0}^{10}.$$

The symbols β, K_0, M_C, M_F denote the four fixed integers of Appendix A. Every other letter in this section is a polynomial abbreviation, not an additional unknown.

2.1 Common arithmetic registers

Define

$$\begin{aligned} h &= \beta J, & q &= h+1, & C &= q-F-\alpha, \\ X &= qw, & Y &= s, & \overline{Y} &= qY, \\ k &= \eta+\zeta, & c &= kY+\eta, & A &= \overline{Y}(X+1)+1, \\ \Delta &= A^2-1, & H &= 4A-5, & v &= X\overline{Y}^2, \\ \ell &= vk, & E &= 1+\ell(\ell+k). \end{aligned} \tag{1}$$

The remaining shared registers are

$$\begin{aligned} R &= [h(q-F)+\alpha](q^2-1)+(M_C+qM_F)J, \\ B &= (K_0+w)C-F, & U &= ic^2, \\ a &= \Delta U^2 E, & F_0 &= 1+a, & G &= EF_0, \\ f &= 1+2a, & Q &= 4\Delta a F_0, & b &= cT-R, \\ V &= fb-c, & K &= X(3c-2X)+2, \\ e_6 &= 6(k-R-1)-\ell(k^2-4), & D &= 6q^7, & M &= hDH. \end{aligned} \tag{2}$$

The subscript on K_0 distinguishes the fixed coefficient from the polynomial K. The numeral 6 in D is part of the literal equation.

2.2 The center and the transport

Set

$$\begin{aligned} e_2 &= (k\overline{Y})^2, \\ S &= (k\overline{Y})\big[\ell+k-(k\overline{Y})(\overline{Y}+1)\big], \\ \theta &= U^2 b, \qquad R_c = \theta(3E+e_2)-(b-c), \\ S_c &= 4E^2+3S^2+e_2^2, \\ \Omega &= U\big[2SR_c+A(4EV-\theta S_c)\big], \\ L &= AU^3b^2. \end{aligned} \tag{3}$$

The useful folded evaluation is

$$\Omega_{\mathrm{f}} = U\big[2SR_c+A\{4EV-\theta(S_c-bB)\}\big]. \tag{F}$$

It satisfies the exact identity

$$\Omega_{\mathrm{f}} = \Omega + LB. \tag{F1}$$

One can therefore compute the carrier directly as

$$Z = hD\big[H(n-\Omega_{\mathrm{f}})+K\big]-H\big[DB+h(e_6-6q^4 s)\big], \tag{4}$$

$$P_{190} = Q(Z^2-16M^2GV^2)+16M^2G-Z^2. \tag{5}$$

$$\boxed{P_{190}(J,F,\alpha,i,T,w,s,n,\eta,\zeta)=0, \quad J,F,\alpha,i,T,w,s,n,\eta,\zeta>0.}$$

Equations (1)–(5), together with the coefficient appendix, specify one ordinary polynomial with integer coefficients. Quotients, modular arithmetic, Pell sequences, and roots used below belong only to the proof. They are not primitives in this polynomial.

3 What is inherited and what changes

Source 172 supplies the free chart $Y = s$, $\overline{Y} = qY$, the inverse-Lucas correction e_6, the denominator $6q^7$, and the signed analysis that permits auxiliary Pell index three. Its complete local proof and the selected underlying proof files are retained in the archive. Source 178 supplies the center for the whole product and its polynomial-quotient identity. The incoming Affine Horizons archive supplies the degree-206 comparison [1, 2, 3].

The new contribution is the positive weight $L = AU^3b^2$, its independent bound in the free chart, the folding identity (F1), and the literal degree-190 equation. These ingredients are needed together. An older unweighted b^2 estimate relied on excluding an auxiliary index that the free chart allows. Importing that exclusion would invalidate the argument.

After root extraction and sign recovery, the inherited Pell authentication and ratio projector recover

$$X = 2^R, \qquad q^3 \mid Y, \qquad C > 0, \qquad \frac{(K_0 + w)C - F}{h} \in \mathbb{N}. \tag{N}$$

The entire calibrated four-number tuple then reconstructs the native words at the source base $Y(X+1)+2$; the inherited native/K215 refutation calibration produces a finite syntactic ZFC refutation. Conversely, a refutation gives canonical positive outer data, compatible Pell witnesses, and the completeness bound used below. Neither direction permits replacing the actual four integers by arbitrary integers satisfying the same size inequalities.

The direct construction in Section 7 uses a different calibration: the numerical C222 calculus, then ZF refutation and ZFC refutation through the named `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation` interfaces. It does not use the four-number tuple.

These are mathematical dependency boundaries. No newly compiled complete Lean/Rocq import closure or object-language ZFC derivation is claimed. The new arguments use integer algebra, finite factorization, induction, and the inherited theorems, and can be carried out internally in ZFC. Finite checks validate emitted arithmetic; they do not prove unbounded semantic assertions by sampling. The archive's `calibration/` folder also records an inspection of the actual unconditional endpoint declarations, their mathematical source dependencies, and selected archived assumption logs. It retains precise source pins and distinguishes those historical logs from a fresh proof-assistant build.

4 Soundness of the degree-190 equation

4.1 Root extraction before sign recovery

At every positive input, $q \geq 32$, $X \geq q$, $Y \geq 1$, $k \geq 2$, $c \geq 3$, $U \geq c^2$, and

$$A \geq 32(32+1)+1 = 1057. \tag{B0}$$

The identities

$$f^2 = 1 + 4aF_0, \qquad Q = \Delta(f^2-1) = 4\Delta^2U^2G, \qquad \gcd(E, F_0) = 1 \tag{I}$$

are independent of the carrier. Since $f > 2c$, $V = fb - c$ is nonzero for every integral b. Suppose $P_{190} = 0$. Rewriting the final norm gives

$$(Q-1)Z^2 = 16M^2G(QV^2-1). \tag{R}$$

A prime dividing G divides neither $Q-1$ nor QV^2-1. Valuations in (R) make G a square. Coprimality in (I) gives positive integers τ, f_0, g with

$$E = \tau^2, \quad F_0 = f_0^2, \quad G = g^2, \quad g = \tau f_0. \tag{R1}$$

The factor 16 changes no valuation parity. The inherited adjacent-coefficient Pell descent applies initially to the rational number $|Z|/(4Mg)$ and proves it integral. Thus

$$QV^2 - (Q-1)y^2 = 1, \qquad Z/M = \epsilon\, 4gy, \qquad y \in \mathbb{Z}_{>0}, \quad \epsilon \in \{-1, 1\}. \tag{R2}$$

One does not assume numerator divisibility in advance.

The inherited signed all-index folding, applied to

$$I = 2Ug, \quad f^2 - \Delta I^2 = 1, \quad Q = (\Delta I)^2, \quad c^2 \mid I, \quad V \equiv -c \pmod{f},$$

gives $c = \mathsf{Y}_p(A)$ for positive odd p. As $c \geq 3$, $p \geq 3$, and

$$c \geq 4A^2 - 1 = 4\Delta + 3, \qquad c > X, \qquad K > 0. \tag{R3}$$

No sign of B, C, R, b, V, and no divisibility of Y by q, has been used. The same parent analysis proves that b, V are nonzero with the same sign, including the auxiliary-index-three case, and

$$|b| > 4\Delta f - 1 > 3\Delta f, \qquad y < \tfrac{3}{2}|V|, \qquad |V| < \tfrac{3}{2}f|b|, \qquad f > 6g. \tag{R4}$$

These signed bounds are the appropriate input to the new argument.

4.2 An absolute bound for the polynomial center

For proof only, replace A by a formal indeterminate t, keeping $\overline{Y}, k, U, b, c$ fixed. Put

$$\begin{aligned} E(t) &= e_2t^2 + e_1t + e_0, \\ e_1 &= k^2\overline{Y}(1 - 2\overline{Y} - 2\overline{Y}^2), \\ e_0 &= 1 + k^2\overline{Y}(\overline{Y}+1)\big[\overline{Y}(\overline{Y}+1) - 1\big], \\ V(t) &= \big[1 + 2(t^2-1)U^2E(t)\big]b - c, \qquad F(t) = E(t)V(t). \end{aligned}$$

Let $\mathcal{Q}_j$ be the ordinary polynomial quotient of $F(t)$ by t^j, evaluated at A. Exact algebra gives

$$\Omega = U\left(4AEV - 2\mathcal{Q}_1 - \frac{\mathcal{Q}_3}{2}\right). \tag{C1}$$

The quotient $\mathcal{Q}_3$ is coefficientwise even: the term $E(t)(b-c)$ has degree at most two and the remaining term has factor $2U^2b$. Equation (C1) is proof notation for the division-free center in (3), not numerical polynomial division in the equation.

Let $F^{\#}(A)$ mean evaluation at positive A after replacing every coefficient by its absolute value. The bound $\delta = (\overline{Y}+1)/A < 1/16$ follows from $\overline{Y}, X \geq 32$. Using $|e_1| < 2e_2(\overline{Y}+1)$ and $0 < e_0 \leq e_2(\overline{Y}+1)^2$ gives

$$E^{\#}(A) < 2E, \qquad F^{\#}(A) < 2E(|b|+c) + 8(A^2+1)U^2E^2|b|.$$

Since $c+1 < 2U^2E$ and $|V| > \Delta U^2 E|b|$,

$$\frac{F^{\#}(A)}{|EV|} < 8 + \frac{20}{\Delta} < 10. \tag{C2}$$

Deletion of low coefficients yields $|\mathcal{Q}_j| \leq F^{\#}(A)/A^j$. Hence

$$|\Omega - 4AUEV| < 10U|EV|\left(\frac{2}{A} + \frac{1}{2A^3}\right) < AUE|V|.$$

We obtain

$$\operatorname{sgn}(\Omega) = \operatorname{sgn}(V), \qquad 3AUE|V| < |\Omega| < 5AUE|V|. \tag{C3}$$

No premature assumption $V > 0$ is hidden in this estimate.

4.3 The weighted transport inequality

Lemma 4.1. *At every positive zero of P_{190},*

$$E < U^2, \qquad 4gy + |\Omega| < \tfrac{1}{2}L, \qquad L = AU^3b^2 > 0.$$

Proof. From $v = X\overline{Y}^2$, $A = \overline{Y}(X+1)+1$, and (R3),

$$v + 1 < A^2 < c, \qquad k < c.$$

These quantities are integers, so

$$E = 1 + k^2 v(v+1) \leq 1 + (c-1)^4 < c^4 \leq U^2.$$

Also $(AUE)^2 - G = U^2E^2 - E > 0$, so $g < AUE$. Combine this with (C3) and (R4):

$$\begin{aligned} 4gy + |\Omega| &< 6g|V| + 5AUE|V| \\ &< 11AUE|V| < \frac{33}{2}AUEf|b| \\ &< \frac{11E}{2\Delta U^2}\,AU^3b^2 < \frac{11}{2\Delta}L < L/2. \end{aligned}$$

The last inequality uses $\Delta > 11$. All estimates precede recovery of the sign of B. □

The weight fits both the signed estimate and a factor already present in the center. Identity (F1) absorbs it into the literal evaluator.

4.4 Recover the arithmetic sign

The following are rational abbreviations used only in the proof:

$$e^* = e_6/6, \quad \rho = e^*/q^7, \quad \sigma = s/q^3, \quad t = B/h, \quad \kappa = K/H, \quad z = Z/M, \quad W_c = 1 + hL.$$

Expansion gives the exact identity

$$z = n + \kappa + \sigma - W_c t - \rho - \Omega. \tag{T}$$

We have $\kappa, \sigma > 0$. Since $k \geq 2$, the inverse correction obeys

$$e^* \leq k - R - 1 < b, \qquad \rho < |b|/q^7. \tag{T1}$$

This remains valid for negative R. If $B \leq -1$, then $W_c t < -L$. As $L \geq |b|$, Lemma 4.1 and (R2) give

$$n = \epsilon 4gy + \Omega - \kappa - \sigma + W_c t + \rho < L/2 - L + |b|/q^7 < 0.$$

This contradicts the independent condition $n > 0$. Thus $B \geq 0$, and then $C > 0$.

4.5 Authentication and native recovery

Only now may one invoke the calibrated mask estimate $3q < R < q^4 - 2$. It gives $b, V > 0$; the unchanged signed index argument yields $p = R$, R odd, q even, and J odd. The carrier changes do not affect $V \equiv -c \pmod f$ or $V \equiv -R \pmod c$.

The center shift $M\Omega$ is an integer multiple of D and H, and L is an integer polynomial. The parent numerator reductions and fractional-part argument therefore remain valid, in this order:

1. The first Pell pair gives $k = 2\mathsf{Y}_m(1 + 2v)$. Literal cancellation of 6 supplies $q^4 \mid e^*$.
2. The retained cubic-difference identity, including its prime-3 valuation argument, yields $m = (R + 1)/2$. One does not yet infer $q^7 \mid e^*$.
3. Reduction modulo H gives $H \mid 6hK$. The $\lambda = q$ ratio theorem and universal projector give $X = 2^R$. Since $X = qw$, q is a power of two.
4. The stronger resulting valuation of $v = Xq^2Y^2$ yields $q^7 \mid e^*$. Now ρ, κ are integral, and (T) gives $q^3 \mid Y$ and $h \mid B$.

These are the calibrated endpoint (N). Fresh native reconstruction produces a finite ZFC refutation. This completes soundness.

5 Completeness and the consistency equivalence

Start with a finite syntactic ZFC refutation. The inherited construction gives canonical positive outer data, compatible positive Pell witnesses, and a positive trace-linear coordinate T, with

$$b, V, y > 0, \quad t = B/h \in \mathbb{N}, \quad \rho, \kappa, \sigma \in \mathbb{Z}, \quad |\rho| + \kappa + \sigma < g. \tag{E}$$

The last bound is proved independently of the carrier weight. Retain every input except n and set

$$n = 4gy + \Omega + (1 + hL)t + \rho - \kappa - \sigma. \tag{E1}$$

It is an integer. Since $\Omega > 0$, $t \geq 0$, and $y \geq 1$, (E) proves it positive. Equation (T) says $Z/M = 4gy$, and the final norm vanishes.

A positive zero therefore exists if and only if a finite syntactic ZFC refutation exists. Negation proves Theorem 1.1. The argument does not assume consistency or assert insolubility.

Remark 5.1. Completeness constructs canonical witnesses. It is not a pointwise bijection of all old and new positive zero sets. The inverse of a polynomial translation of a positive coordinate need not remain positive. The separate signed soundness argument is essential.

6 Exact degree and a cheaper companion

6.1 The degree-190 certificate

Let Z_0 be the carrier with transport L, before subtraction of $M\Omega$. Thus $Z = Z_0 - M\Omega$. Regard $(A, \overline{Y}, k, U, b, c)$ as free variables of weights $(4, 2, 1, 5, 4, 2)$. Exact collection gives 594 nonzero terms in the unfolded center defect, weighted degree 94, and top part

$$[\Omega^2 - 16GV^2]_{94} = 8A^6\overline{Y}^8k^8U^6b^2(A^2 - 6\overline{Y}^4). \tag{D1}$$

Using the folded center instead gives 679 terms, with the same top part; its additional transport terms have lower weight. In the ten actual inputs,

$$\deg(Q, M, G, V, \Omega, Z_0) = (72, 12, 46, 36, 59, 41).$$

The cancellation-sensitive expansion is

$$P_{190} = Q\{Z_0^2 - 2MZ_0\Omega + M^2(\Omega^2 - 16GV^2)\} + 16M^2G - (Z_0 - M\Omega)^2. \tag{D2}$$

The five degree bounds are $154, 184, 190, 70, 142$; only the third term can contribute in degree 190. Put $q_0 = \beta J$, $k_0 = \eta + \zeta$. Substitution in (D1) gives the entire top homogeneous form

$$[P_{190}]_{190} = 9\,2^{11}q_0^{74}w^{18}s^{52}i^{10}k_0^{32}(q_0 - F)^2(w^2 - 6s^2). \tag{D3}$$

It is nonzero for the actual $\beta > 0$, proving exact degree. The literal graph specifies the whole polynomial without expansion.

6.2 A 99-operation companion

Keep the common definitions (1)–(2) and S. For this alternative only, put

$$L = bV, \quad C_2 = 2AUE - US, \quad Z = hD[H\{n - V(bB + C_2)\} + K] - H[DB + h(e_6 - 6q^4s)]$$

and

$$P_{202} = Q(Z^2 - 4M^2GV^2) + 4M^2G - Z^2. \tag{C202}$$

These definitions replace the principal carrier; there remain ten positive inputs. The positive remainder

$$R_E = 1 + k^2\overline{Y}(\overline{Y} + 1)\{\overline{Y}(\overline{Y} + 1) - 1\}$$

satisfies $E = AS + R_E$, $0 < S < E/A$, and $0 < C_2 < 2AUE < 3g$. The inherited bound gives $|b| > 16g$, so

$$2gy + C_2|V| < 6g|V| < L/2.$$

The same bootstrap and completeness construction work with root scale two. The identity

$$C_2^2 - 4G = 4U^2ER_E + U^2S^2 - 4E$$

has degree 34. The output has exact degree 202, with top form

$$[P_{202}]_{202} = 9\,2^{12}q_0^{80}w^{22}s^{56}i^{10}k_0^{32}(q_0 - F)^2.$$

Its evaluator uses 60 multiplications, 21 additions, and 18 subtractions.

7 Reusing a Pell quotient as a center witness

The preceding construction retains a native coefficient calibration. The following mechanism operates in the direct C222 proof-calculus branch and removes a private witness by changing how a large center is chosen. Use standard Pell polynomials

$$(a + \sqrt{a^2-1})^j = \mathsf{X}_j(a) + \mathsf{Y}_j(a)\sqrt{a^2-1}, \qquad a \geq 2, \quad j \geq 0.$$

Their recurrences and addition laws are polynomial identities.

Theorem 7.1 (The quotient supplies the center)**.** *Let $q, \mathcal{M}$ be positive odd integers with* $\mathrm{rad}(\mathcal{M}) \mid q$*, and let $W_0 \geq 2$. Set*

$$W = q^2W_0, \quad \mathcal{K} = W^3(W+2), \quad A_0 \equiv 1 + \frac{q^2}{2(q+1)} \pmod{\mathcal{M}}.$$

There are arbitrarily large positive odd j for which

$$L_0 = \mathsf{X}_{Wj}(W+1), \quad r = \frac{\mathsf{Y}_{Wj}(W+1)}{W}, \quad A_* = L_0^2 + 2r^2$$

satisfy

$$r \in \mathbb{Z}_{>0} \text{ is odd}, \quad L_0^2 - \mathcal{K}r^2 = 1, \quad A_* \equiv A_0 \pmod{\mathcal{M}}.$$

In particular $A_* \geq L_0^2$, and these centers are cofinal.

The quotient $r = 2z + 1$ is already a source witness. No extra coordinate stores j, a modular root, or a CRT choice.

7.1 A local unit and a permutation of indices

Lemma 7.2. *If p is odd and $p^2 \mid W$, then*

$$b_0 = \mathsf{Y}_W(W+1)/W \in \mathbb{Z}, \quad b_0 \equiv 1 \pmod{p}, \quad c_0 = \mathsf{X}_W(W+1) \equiv 1 \pmod{p^6}.$$

Proof. The recurrence gives $\mathsf{Y}_j(W+1) \equiv j \pmod{W}$. With $a = W+1$, $d = W(W+2)$, binomial expansion gives

$$b_0 = \sum_{2t+1 \leq W} \frac{\binom{W}{2t+1}}{W} a^{W-2t-1}d^t.$$

The $t = 0$ term is 1 modulo p. For $t \geq 1$,

$$\nu_p\left(\binom{W}{2t+1}/W\right) \geq -\nu_p(2t+1).$$

Each whole term has valuation at least $2t - \nu_p(2t+1) \geq 1$. Although the displayed coefficients can be rational, each whole term is p-integral, and these valuation bounds are legitimate. Thus $b_0 \equiv 1 \pmod{p}$. Also $c_0 \equiv 1 \pmod{p}$. The norm identity $c_0^2 - 1 = dW^2 b_0^2$ has right side divisible by p^6, while $c_0 + 1$ is a unit. This proves the claim. □

Lemma 7.3 (Index permutation)**.** *If p is odd and $c_0 \equiv 1 \pmod{p^2}$, then, for $a, j \geq 0$,*

$$\mathsf{Y}_{j+p^a}(c_0) - \mathsf{Y}_j(c_0) \equiv p^a \pmod{p^{a+1}}. \tag{Perm}$$

Hence $j \mapsto \mathsf{Y}_j(c_0)$ is a permutation modulo every power of p.

Proof. Put $h = p^a$, $d' = c_0^2 - 1$. The elementary bound $\nu_p\binom{h}{t} \geq a - \nu_p(t)$ and $\nu_p(d') \geq 2$ give

$$\mathsf{X}_h(c_0) \equiv 1 \pmod{p^{a+1}}, \qquad \mathsf{Y}_h(c_0) \equiv p^a \pmod{p^{a+1}}.$$

Every nonleading ordinate term has valuation at least $a + 2t - \nu_p(2t+1) \geq a + 1$. The scalar expansion is similar; its leading power c_0^h is 1 modulo p^{a+1}. The addition law

$$\mathsf{Y}_{j+h}(c_0) = \mathsf{X}_h(c_0)\mathsf{Y}_j(c_0) + \mathsf{X}_j(c_0)\mathsf{Y}_h(c_0)$$

proves (Perm), using $\mathsf{X}_j(c_0) \equiv 1 \pmod{p}$. Modulo p the map is the identity. At the next lift, adding tp^a changes the new base-p digit by t. Induction proves bijectivity. □

7.2 Proof of the center theorem

If $\mathcal{M} = 1$, every odd positive j suffices. Otherwise $p^2 \mid W$ for each $p \mid \mathcal{M}$. The composition law

$$r(j) = \mathsf{Y}_{Wj}(W+1)/W = b_0 \mathsf{Y}_j(c_0)$$

and the lemmas make $j \mapsto r(j)$ a permutation modulo every prime power dividing $\mathcal{M}$. CRT gives a permutation modulo $\mathcal{M}$. For each such prime,

$$2(q+1)(\mathcal{K}+2) \equiv 4 \pmod{p}.$$

The simple roots $2, -2$ lift to all required prime powers by $d \mapsto d + tp^a$, since $2d$ is invertible. CRT supplies a unit $d \pmod{\mathcal{M}}$ with

$$d^2 \equiv 2(q+1)(\mathcal{K}+2) \pmod{\mathcal{M}}.$$

Choose j with $r(j) \equiv q/d \pmod{\mathcal{M}}$. Because $\mathcal{M}$ is odd, this class has odd positive representatives; adding $2\mathcal{M}$ gives arbitrarily large ones. The norm identity now gives

$$A_* = 1 + (\mathcal{K}+2)r^2 \equiv 1 + \frac{q^2}{2(q+1)} \pmod{\mathcal{M}}.$$

For odd W, $\mathsf{Y}_W(W+1)/W$ is odd by reduction modulo two. For even W, all abscissae at $W+1$ are odd, and $\mathsf{Y}_{2t} = 2\mathsf{X}_t\mathsf{Y}_t$ proves $\nu_2(\mathsf{Y}_W) = \nu_2(W)$. The quotient is again odd. Finally $\mathsf{Y}_j(c_0)$ is odd for odd j, regardless of c_0's parity. Positivity and cofinal growth follow from the positive recurrence.

7.3 Why this identifies a new pair of witness roles

The incoming cofinal-center archive already proves that an even cofinal frame permits a separate witness $A_* = L_0^2 + 2u^2$ [5]. Here the existing r replaces u. The changed frame lets that quotient cover the required residue classes. The hypothesis matters: at $W = 3, p = 3$,

$$\Upsilon_3(4)/3 = 21$$

is not a local unit. Requiring merely $p \mid W$ would not justify the proof.

In package 166 take $W_0 = 2 + \sum a_i^2$, using exactly its old affine atoms. Replace

$$W - W_0 = 0, \qquad A_* - L_0^2 - u^2 - v^2 = 0$$

by

$$W - q^2 W_0 = 0, \qquad A_* - L_0^2 - 2(2z+1)^2 = 0. \tag{Reuse}$$

Delete the now private u, v; all other source rows are unchanged. Soundness first uses the affine radix row $q = 6t + 3$, hence $W = q^2 W_0 \geq W_0$, and $A_* \geq L_0^2$. It does not assume $\mathcal{M} \neq 0$ before power recovery. Completeness first constructs the semantic positive odd $q, \mathcal{M}$ with the required prime support, and then applies Theorem 7.1.

This is an equivalence of zero existence, not a map from every preselected old center. The raw frame row has degree four, giving a degree-eight sum of squares. A signed-frame refinement and exact elimination give the stronger direct endpoints below.

8 A direct equation in 83 unrestricted integers

The modular witness-reuse theorem leads to a concrete second construction. Its source is the unmodified direct numerical-calculus polynomial in package 166, followed by the two semantic changes just proved and explicit integral coordinate elimination. The main endpoint is

$$D_{83}(x_0, \ldots, x_{82}) = \sum_{j=0}^{30} R_j(x_0, \ldots, x_{82})^2. \tag{Direct}$$

Appendix C prints every free assignment, every ordered substitution, and every residual. The electronic appendix also supplies the corresponding literal arithmetic circuit and source coefficients. All 83 inputs range independently over $\mathbb{Z}$. There are no hidden inequalities or extra witnesses.

Theorem 8.1 (Direct C222 route)**.** *For the fixed polynomial (Direct), with the inherited C222/ZF/ZFC refutation calibration identified in Section 3,*

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{83}\ D_{83}(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

The exact total degree is 624*, and the preferred factored evaluator has* 1,789 *live binary arithmetic operations. Its integer literal scalars have absolute value at most* 64*.*

This route does not simulate K215 or specialize Jones's universal polynomial. It retains a specific direct proof calculus and its previously identified consistency calibration. Compared with the 2,428-operation 92-integer, degree-692 equation [7], the new point improves dimension $92 \to 83$, degree $692 \to 624$, and operation count $2{,}428 \to 1{,}789$ together. The latter count uses the exact factorizations described below; the canonical evaluator retained for the projection audit has 2,939 operations.

8.1 A signed frame lowers the source degree

For the source used in this endpoint, one may replace q^2W_0 by qW_0. During completeness, the semantic construction gives $q = (BH)^x$, with $x \geq 2$, and every prime dividing the odd modulus $\mathcal{M} = Nq^x$ divides q. Thus every such prime has square dividing $W = qW_0$. The proof of Theorem 7.1 only needs that local square divisibility, so the same modular construction works. The positivity of the semantic q gives a positive completeness frame.

Soundness must also cover a negative value of the unclassified input q. The affine source row gives $|q| \geq 3$, so $F = |W| \geq 3W_0$. The new $A_* = L_0^2 + 2r^2$ is positive and at least L_0^2. When $W = -F$, the frame equation becomes

$$L_0^2 - F^3(F-2)r^2 = 1.$$

This is the Pell equation at parameter $F - 1$, with ordinate $F|r|$. The congruence

$$\mathsf{Y}_j(F-1) \equiv (-1)^{j-1}j \pmod{F}$$

forces its positive classified index to be a multiple of F. As r is odd, it is nonzero, so that index is at least F. Consequently

$$A_* \geq L_0^2 \geq (F-1)^{2F} > F^F \qquad (F \geq 3).$$

For $W > 0$ the retained positive-frame argument gives the required bound. Every old bounded atom has absolute value below F. The source's only uses of W and its private frame wires are the frame equations and its definition, checked by an occurrence audit. The inherited signed power recognition therefore applies with $F = |W|$.

The signed-frame row has degree three, rather than four. Replaying the inherited integral coordinate changes, with the two private center inputs deleted, gives an explicit equation in 227 integers, the sum of 175 residual squares, of exact degree six. Its collected polynomial has 32,328 monomials and height 221,184. The archive also retains the positive-frame degree-eight alternative. The 227-variable intermediate is a proof source, not a global quartic or low-dimensional record.

8.2 Acyclic unit elimination

Let $S_0, \ldots, S_{174}$ be the literal residuals in that 227-variable source. A selected row has the form

$$S_r = \epsilon x_v + H, \qquad \epsilon \in \{1, -1\}, \qquad H \in \mathbb{Z}[x_0, \ldots, \widehat{x_v}, \ldots, x_{226}]. \tag{Pivot}$$

Set $x_v = -\epsilon H$ and remove that row. The dependence graph of the selected definitions is acyclic, so the removed coordinates are restored as integer polynomials in the retained ones, in a finite order. No division by a nonunit is performed.

For the principal endpoint the certificate contains 144 such definitions, leaving 83 coordinates and 31 residuals. If $\Phi : \mathbb{Z}^{83} \to \mathbb{Z}^{227}$ is the resulting reconstruction, exact polynomial substitution gives

$$S_r(\Phi(\boldsymbol{x})) = 0 \quad \text{for every removed row,} \qquad D_{227}(\Phi(\boldsymbol{x})) = D_{83}(\boldsymbol{x}). \tag{Restore}$$

Projection and reconstruction are inverse on the integer zero sets. The signed-frame and witness-reuse changes earlier preserve existence; the acyclic elimination itself gives this stronger zero-set bijection. The method of unit elimination is inherited from the repository's prior projection work. The new semantic source, selected dependency orientation, literal certificates, and measured endpoints are supplied here.

8.3 Exact degree and cost certificates

The circuit gives an upper bound 312 on each retained residual degree. A univariate specialization $x_i = d_i t$, with explicitly supplied integer d_i, has a nonzero t^{312} coefficient in a retained residual modulo 1,000,003. This proves that residual has exact degree 312 over $\mathbb{Z}$. The sum of squares has degree 624: highest real homogeneous squares cannot cancel. Independent finite-field nonzero partial derivatives certify that every one of the 83 declared inputs actually occurs.

Exact propagation of coefficient-norm upper bounds through the circuit gives the rigorous bound

$$\max |\text{expanded coefficient of } D_{83}| < 2^{1568}.$$

This is a bound, not an exact expanded height. A literal scalar bound of 64 in the factored evaluator is a different measure. The complete expanded degree-624 polynomial was not collected; the circuit is its exact finite definition.

8.4 A factored evaluator for the same polynomial

The expanded source-coordinate presentation contains two long expressions that share a quadratic form. Factoring these expressions changes neither the source, the projection, nor the polynomial. It gives a second literal evaluator with **1,789 live binary operations**, comprising 955 multiplications and 834 additions. The largest absolute literal is 64. The canonical evaluator has 2,939 operations and largest literal 324. Both define exactly the same polynomial in 83 unrestricted integers, of degree 624, with 31 residual squares and expanded coefficient height below 2^{1568}.

Use the source coordinates $z_0, \dots, z_{226}$ from the projection appendix, and introduce the following polynomial abbreviations:

$$\begin{aligned}
H_* &= z_8 + z_9 + 4(z_{19} + z_{24} + z_{30} + z_{36} + z_{65}) \\
&\quad + 2(z_{22} + z_{27} + z_{29} + z_{32} + z_{34} + z_{37} + z_{40} + z_{44} + z_{66}), \\
q_* &= 3 + 6z_{220}, \\
\mathcal{I} &= \{1, 2, 10, 11, 12, 13, 14, 15, 16, 17, 103, 121, 127, 195\}, \\
\mathcal{J} &= \{145, 153, 158, 161, 164, 167, 170, 173, 177, 180, 183, 187, 190, 214, 215, 216\}, \\
Q_* &= z_0^2 + (z_0 + z_4)^2 + \sum_{j \in \mathcal{I}} z_j^2, \\
N_* &= z_{193}^2 + 2z_{158} - z_{216} - \sum_{j \in \mathcal{J}} z_j^2.
\end{aligned}$$

The long restoration of z_{139} has the exact factorization

$$\begin{aligned}
z_{139} &= q_*\big[Q_* - 2z_8 + 4z_{225}(z_{225} + 1) + q_*^2 + 4\big] \\
&\quad + 2(1 + 3z_{220})(H_*^2 + z_8^2 + z_9^2) - 2z_8 z_9 + z_{19} + N_* - 3.
\end{aligned} \tag{6}$$

The corresponding retained residual is

$$R_{28} = N_* - 1 - z_{19} - H_*^2 + (z_8 + z_9)^2. \tag{7}$$

Here residual indexing is zero based: R_{28} is retained source row 170. These letters denote expression abbreviations, not additional unknowns.

The independent checker expands (6) and (7) in the free polynomial ring $\mathbb{Z}[z_0, \dots, z_{226}]$ and compares all 336 and 154 nonzero coefficients, respectively, with the frozen definitions. It then checks every

restoration, every retained residual, and the sum-of-squares output in both complete arithmetic DAGs. Thus the equality holds at every integer tuple, not only on their zero sets. The original degree and occurrence certificates also replay for the alternate evaluator. Exact propagation of coefficient ℓ_1 majorants again gives the strict bound 2^{1568}; this is distinct from the largest literal 64.

The alternate graph, formulas, byte-reproducible builder, independent standard-library checker, and receipt are in `direct/factored83/`. The canonical graph and certificates remain in `direct/P83/`. No semantic theorem is changed by this evaluator optimization.

Integer variables	Exact degree	Residual squares	Interpretation
79	8,388,620	27	Fewest coordinates in this family
81	2,048	29	Smaller dimension, higher degree
82	764	30	Intermediate point
83	**624**	**31**	**Principal direct endpoint**
87	268	35	Lower-degree alternative
89	128	37	Lower-degree alternative
91	80	39	Lower-degree alternative

Each row is an independently specified ordinary polynomial over unrestricted integers, with its own literal circuit and exact degree certificate. The 79-variable equation improves on the 87-variable coordinate count in the inspected oriented-power family, at a slightly higher degree than its 8,388,612 endpoint [6]. The table makes the degree cost explicit. It is not a claim that an equation in 79 variables is simpler than every equation in fewer variables using the large native coefficient tuple.

9 Verification, reproducibility, and remaining work

The archive distinguishes three kinds of evidence:

1. **Literal arithmetic.** JSON circuits and residual lists specify every polynomial, input, fixed coefficient, and operation. The four native integers are supplied as bytes and by a finite reconstruction.
2. **Finite certificates.** Exact sparse algebra checks the center identity, its norm defect, highest forms, and gate counts. Separate code checks source edits, acyclic unit eliminations, reconstruction identities, and the supplied collected coefficients. Finite Pell examples exercise the lemma and a counterexample; they are diagnostics, not the proof of the theorem.
3. **Unbounded reasoning.** The signed transport, index permutation, cofinal completeness, and inherited refutation calibrations are ordinary mathematical proofs. No finite search or successful test suite establishes these universal statements by itself.

Run `verify.py` for the release checks and `build_report.sh` to compile the report. The single-file `report.tex` assembles the readable article and coefficient specification. Electronic appendices contain the direct polynomials, complete restoration data, source snapshots, and receipts. The finite code does not search for an actual refutation or integer zero.

The results give concrete new equations and a reusable way to identify two witness roles. Formalization of the new bridge end to end, and further improvements under a specified simplicity objective, remain open. No worldwide novelty priority is asserted. The independent reviews were performed by separately assigned AI agents and the primary assistant, not by external human peer reviewers.

A The complete four-integer coefficient specification

This is the retained directly indexed BFS943/radix67 tuple. It makes P_{190} and P_{202} fully specified without an unspecified machine, oracle coefficient, or consistency-dependent search. Finite sums, floors, and the fixed seed below define four constants once and for all. They do not add primitives to the polynomial in ten variables.

In this appendix K denotes the port K_0 in the new equations. Every symbol introduced here is a fixed constant, independent of the ten quantified variables. The formulas and 436-character seed are the retained indexed BFS943 radix67 specification; they reconstruct the exact supplied payloads, with $\beta = 2^{590069} - 1$.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} = {} & \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ & (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ & (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j)l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$432 + b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434 + b\ (b = 0, 1)$	$(0, b, 0, b)$
$424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q, b))$
$424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q, b))$
$924 + j\ (0 \le j < 4)$	$(0, j + 430\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934+3a+b-6\lfloor a/4 \rfloor$ $a \in \{0,1,4\},\ b=0,1,2$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is:

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
OOe35uwDf15DpT6VJWZBjSbHPpPAINYTteOeUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

B Literal ten-variable circuit

The following generated table prints the principal circuit in order. Names denote local polynomial registers; the four fixed ports do not count as inputs. Every row is one binary operation, and all 118 rows are live.

Row	Register	Definition
1	bJ	beta · J
2	q	bJ + 1
3	q2	q · q
4	q4	q2 · q2
5	q6	q4 · q2
6	q7	q6 · q
7	sixq7	6 · q7
8	qY	q · s
9	qmF	q − F
10	C	qmF − alpha
11	X	w · q
12	k	eta + zeta
13	kY	k · s
14	c	kY + eta
15	XY	X · qY
16	v0	XY · qY
17	Am1	XY + qY
18	4Am1	4 · Am1
19	Hp	4Am1 − 1
20	Ap1	Am1 + 2
21	V0k	v0 · k
22	V0kk	V0k + k
23	V0k2	V0k · V0kk
24	Dtau	V0k2 + 1
25	G1	bJ · qmF
26	G	G1 + alpha
27	q2m1	q2 − 1
28	Gq	G · q2m1
29	qMF	q · MF
30	MCqMF	MC + qMF
31	RJ	MCqMF · J
32	R	Gq + RJ
33	Rp1	R + 1
34	e	k − Rp1
35	ksq	k · k
36	ksqm4	ksq − 4
37	inverse_cubic	V0k · ksqm4
38	q4s	q4 · s
39	e_sigma	e − q4s
40	six_e_sigma	6 · e_sigma
41	eclear	six_e_sigma − inverse_cubic
42	K0w	K0 + w
43	K0wC	K0w · C
44	Bt	K0wC − F
45	c2	c · c
46	U	i · c2
47	UE	U · Dtau
48	Delta	Am1 · Ap1
49	SE	Delta · UE
50	DpI2	SE · U
51	Df	DpI2 + 1
52	aF0	DpI2 · Df
53	fouraF0	4 · aF0
54	trace_f	DpI2 + Df

Row	Register	Definition
55	Q	Delta · fouraF0
56	cT	c · T
57	cTR	cT − R
58	DfcTR	trace_f · cTR
59	V	DfcTR − c
60	V2	V · V
61	P1mu2	bJ · sixq7
62	new_A	Am1 + 1
63	new_kZ	k · qY
64	new_Zp1	qY + 1
65	new_kZZp1	new_kZ · new_Zp1
66	new_Sinner	V0kk − new_kZZp1
67	new_S	new_kZ · new_Sinner
68	new_e2	new_kZ · new_kZ
69	new_U2	U · U
70	new_theta	new_U2 · cTR
71	new_threeE	3 · Dtau
72	new_3Ee2	new_threeE + new_e2
73	new_theta3Ee2	new_theta · new_3Ee2
74	new_bmc	cTR − c
75	new_Rc	new_theta3Ee2 − new_bmc
76	new_E2	Dtau · Dtau
77	new_fourE2	4 · new_E2
78	new_S2	new_S · new_S
79	new_threeS2	3 · new_S2
80	new_e22	new_e2 · new_e2
81	new_sc1	new_fourE2 + new_threeS2
82	new_Sc	new_sc1 + new_e22
83	new_bB	cTR · Bt
84	new_ScShift	new_Sc − new_bB
85	new_EV	Dtau · V
86	new_fourEV	4 · new_EV
87	new_thetaSc	new_theta · new_ScShift
88	new_Vdiff	new_fourEV − new_thetaSc
89	new_Apart	new_A · new_Vdiff
90	new_SRc	new_S · new_Rc
91	new_twoSRc	2 · new_SRc
92	new_omegainner	new_Apart + new_twoSRc
93	new_Omega	U · new_omegainner
94	center	n − new_Omega
95	Hn	Hp · center
96	threec	3 · c
97	twoX	2 · X
98	gap	threec − twoX
99	Kpos0	X · gap
100	Kpos	Kpos0 + 2
101	lin	Hn + Kpos
102	Zpart1	P1mu2 · lin
103	mu2inner	sixq7 · Bt
104	P1e	bJ · eclear
105	brk	mu2inner + P1e
106	Zpart2	Hp · brk
107	Znum	Zpart1 − Zpart2
108	Mtot	P1mu2 · Hp
109	M2	Mtot · Mtot
110	Dg	Dtau · Df
111	new_scaledG	16 · Dg
112	Z2	Znum · Znum
113	RG	M2 · new_scaledG
114	RGV2	RG · V2
115	inside	Z2 − RGV2

Row	Register	Definition
116	`Qinside`	`Q` · `inside`
117	`correction`	`RG` − `Z2`
118	`N`	`Qinside` + `correction`

C Complete finite definition of the direct polynomial

The only independent unknowns in this appendix are $x_0, \dots, x_{82} \in \mathbb{Z}$. Every z_i below is an abbreviation for an integer polynomial in these unknowns. First make the free-coordinate assignments, then evaluate the acyclic definitions in their printed order. Finally evaluate the 31 residuals and add their squares. This is a complete ordinary polynomial definition using fixed integer coefficients, addition, and multiplication.

Free-coordinate assignments

$$
\begin{aligned}
z_5 &= x_0, & z_6 &= x_1, & z_8 &= x_2, & z_9 &= x_3 \\
z_{10} &= x_4, & z_{11} &= x_5, & z_{12} &= x_6, & z_{13} &= x_7 \\
z_{14} &= x_8, & z_{15} &= x_9, & z_{18} &= x_{10}, & z_{48} &= x_{11} \\
z_{53} &= x_{12}, & z_{62} &= x_{13}, & z_{72} &= x_{14}, & z_{101} &= x_{15} \\
z_{104} &= x_{16}, & z_{109} &= x_{17}, & z_{111} &= x_{18}, & z_{112} &= x_{19} \\
z_{115} &= x_{20}, & z_{118} &= x_{21}, & z_{121} &= x_{22}, & z_{122} &= x_{23} \\
z_{124} &= x_{24}, & z_{125} &= x_{25}, & z_{128} &= x_{26}, & z_{131} &= x_{27} \\
z_{132} &= x_{28}, & z_{134} &= x_{29}, & z_{136} &= x_{30}, & z_{137} &= x_{31} \\
z_{141} &= x_{32}, & z_{144} &= x_{33}, & z_{145} &= x_{34}, & z_{147} &= x_{35} \\
z_{149} &= x_{36}, & z_{151} &= x_{37}, & z_{152} &= x_{38}, & z_{153} &= x_{39} \\
z_{156} &= x_{40}, & z_{157} &= x_{41}, & z_{158} &= x_{42}, & z_{161} &= x_{43} \\
z_{163} &= x_{44}, & z_{164} &= x_{45}, & z_{166} &= x_{46}, & z_{167} &= x_{47} \\
z_{169} &= x_{48}, & z_{170} &= x_{49}, & z_{173} &= x_{50}, & z_{175} &= x_{51} \\
z_{177} &= x_{52}, & z_{179} &= x_{53}, & z_{180} &= x_{54}, & z_{183} &= x_{55} \\
z_{185} &= x_{56}, & z_{187} &= x_{57}, & z_{189} &= x_{58}, & z_{190} &= x_{59} \\
z_{193} &= x_{60}, & z_{196} &= x_{61}, & z_{197} &= x_{62}, & z_{202} &= x_{63} \\
z_{203} &= x_{64}, & z_{204} &= x_{65}, & z_{205} &= x_{66}, & z_{206} &= x_{67} \\
z_{210} &= x_{68}, & z_{211} &= x_{69}, & z_{212} &= x_{70}, & z_{213} &= x_{71} \\
z_{214} &= x_{72}, & z_{215} &= x_{73}, & z_{216} &= x_{74}, & z_{217} &= x_{75} \\
z_{218} &= x_{76}, & z_{219} &= x_{77}, & z_{220} &= x_{78}, & z_{222} &= x_{79} \\
z_{223} &= x_{80}, & z_{225} &= x_{81}, & z_{226} &= x_{82}
\end{aligned}
$$

Acyclic unit definitions

$$
\begin{aligned}
z_0 &= 16 + z_{196}^2 \\
z_{209} &= z_{193}^2 \\
z_{198} &= 2z_{203}z_{209} + z_{209} \\
z_{201} &= 4z_{198} + 8z_{198}z_{202} \\
z_{140} &= -6 - z_{197}z_{204} + z_{201}
\end{aligned}
$$

$$
\begin{aligned}
z_{146} &= z_{140}z_{145} + 7z_{145}\\
z_{4} &= 1 - 32z_{140}z_{147} + z_{144} + 16z_{145} - z_{146} + 33z_{147}\\
z_{3} &= z_{4} + 2z_{4}z_{225}
\end{aligned}
$$

$$
\begin{aligned}
z_{221} &= 1 + 4z_{222} + 4z_{222}^2\\
z_{2} &= 1 + z_{3}z_{221}\\
z_{7} &= -z_{0} + z_{0}z_{2} + z_{2}\\
z_{1} &= 1 + z_{6}z_{7}\\
z_{16} &= -1 - z_{4} + z_{18} + 2z_{18}z_{225}\\
z_{148} &= 12 + 2z_{140} + 4z_{140}z_{225} + 24z_{225} - 4z_{225}^2\\
z_{224} &= 3 + 2z_{0} + 2z_{4} - 2z_{8} + z_{8}^2 + z_{9}^2 + z_{10}^2 + z_{11}^2 + z_{12}^2 + z_{13}^2 + z_{14}^2 + z_{15}^2\\
&\quad + z_{16}^2\\
z_{195} &= z_{224} + 2z_{224}z_{225}
\end{aligned}
$$

$$
\begin{aligned}
z_{150} &= -1 + 2z_{140}z_{195} + 14z_{195} - z_{195}^2\\
z_{184} &= -1 + z_{140}z_{183} + 7z_{183}\\
z_{182} &= 3 + z_{150}z_{151} - z_{183}z_{195} + z_{184} + 6z_{220}\\
z_{17} &= -z_{148}z_{149} + z_{182} + z_{183} + 2z_{183}z_{225} - z_{184}\\
z_{19} &= 1 + z_{8}^2 + 2z_{8}z_{9} + z_{9}^2\\
z_{20} &= 10z_{8} - z_{8}^2 + 6z_{8}z_{9} + 8z_{8}z_{19} + 20z_{9} + 11z_{9}^2 + 16z_{9}z_{19} + 15z_{19} + 4z_{19}^2\\
z_{21} &= 2 - 2z_{8} + z_{8}^2 + 4z_{8}z_{20} + 4z_{20} + 4z_{20}^2\\
z_{22} &= -1 - 6z_{8} + 36z_{8}^2 + 24z_{8}z_{21} - 2z_{21} + 4z_{21}^2
\end{aligned}
$$

$$
\begin{aligned}
z_{23} &= 2 + z_{9} + z_{9}^2 + 2z_{9}z_{10} + z_{10} + z_{10}^2\\
z_{46} &= 1 + 8z_{10} + 4z_{10}^2 - 8z_{10}z_{23} - 8z_{23} + 4z_{23}^2\\
z_{24} &= 8 - 7z_{8} + z_{8}^2 - 4z_{8}z_{10} + 4z_{8}z_{23} + 4z_{10} - 4z_{23} + z_{46}\\
z_{25} &= 1 - z_{8} + z_{8}^2 + 2z_{8}z_{10} - z_{10} + z_{10}^2\\
z_{31} &= -19 + 25z_{8} + 7z_{8}^2 + 8z_{8}z_{9} + 30z_{8}z_{10} + 16z_{8}z_{25} - 3z_{9} - 5z_{9}^2 + 4z_{9}z_{25} + 27z_{10}\\
&\quad + 21z_{10}^2 + 24z_{10}z_{25} + 6z_{23} + 17z_{25} + 4z_{25}^2\\
z_{26} &= -23 - 16z_{8} + 8z_{8}z_{9} + 8z_{8}z_{10} + 16z_{8}z_{19} + 8z_{8}z_{25} - 19z_{9} - 9z_{9}^2 + 36z_{9}z_{10}\\
&\quad + 12z_{9}z_{25} - 16z_{10} - 8z_{10}^2 + 24z_{10}z_{19} + 8z_{19}z_{25} + z_{20} + z_{31}\\
z_{27} &= 4z_{24}^2 + 8z_{24}z_{26} + 4z_{26}^2\\
z_{28} &= 2 + 8z_{8} - 8z_{8}z_{23} - 8z_{8}z_{25} + 8z_{9} - 8z_{9}z_{23} - 8z_{9}z_{25} + 4z_{19} - 8z_{23} + 4z_{23}^2\\
&\quad + 8z_{23}z_{25} - 8z_{25} + 4z_{25}^2
\end{aligned}
$$

$$
\begin{aligned}
z_{29} &= -6 - 4z_{8} - 32z_{8}z_{9} - 16z_{8}z_{19} - 16z_{8}z_{25} - 8z_{8}z_{28} - 24z_{9} + 32z_{9}z_{25} + 16z_{9}z_{28}\\
&\quad - 12z_{19} + 16z_{19}z_{25} + 8z_{19}z_{28} + z_{20} - 8z_{25} + 16z_{25}^2 + 16z_{25}z_{28} - 4z_{28} + 4z_{28}^2\\
z_{30} &= -22 + 5z_{8} + z_{8}^2 + 8z_{8}z_{9} + 12z_{8}z_{10} + 4z_{8}z_{23} + 8z_{9} + 16z_{9}z_{10} + 16z_{9}z_{23} + 20z_{10}\\
&\quad + 20z_{10}^2 + 24z_{10}z_{23} + 28z_{23} + 4z_{23}^2
\end{aligned}
$$

$$z_{32} = -2 + 4z_{30}^2 + 8z_{30}z_{31} + 4z_{31}^2$$

$$z_{33} = 19z_8 - 8z_8^2 - 8z_8z_{19} - 8z_8z_{23} + 46z_9 + 24z_9^2 - 24z_9z_{19} - 24z_9z_{23} + 2z_{10} - 4z_{19} + 4z_{19}^2 + 8z_{19}z_{23} - 14z_{23} + 4z_{23}^2 + z_{25}$$

$$z_{34} = 3 + 6z_8 + 4z_{10} + 2z_{25} + 4z_{33} + 4z_{33}^2$$

$$z_{38} = 6 + 2z_8 + 4z_9 + 14z_{19} + 16z_{19}^2$$

$$z_{35} = -8 - 42z_8 - 7z_8^2 - 28z_8z_{20} - 48z_9 + 32z_9^2 + 32z_9z_{19} - 64z_9z_{20} + 40z_{19}^2 - 32z_{19}z_{20} - 14z_{20} + 4z_{20}^2 - 2z_{38}$$

$$z_{36} = -1 - 6z_8 + 36z_8^2 + 24z_8z_{35} - 2z_{35} + 4z_{35}^2$$

$$z_{37} = z_9^2 + 4z_9z_{36} + 4z_{36}^2$$

$$z_{39} = -24 - 21z_9 + 21z_9^2 + 18z_9z_{10} + 24z_9z_{23} + 24z_9z_{38} - 3z_{10} + 9z_{10}^2 + 16z_{10}z_{38} - 3z_{23} + 12z_{23}^2 + 8z_{23}z_{38} - 4z_{38} + 4z_{38}^2 - 2z_{46}$$

$$z_{40} = -16 + 10z_8 + 20z_8^2 + 16z_8z_{10} + 24z_8z_{25} + 24z_8z_{39} + 12z_{10} + 16z_{10}z_{25} + 16z_{10}z_{39} + 14z_{25} + 4z_{25}^2 + 8z_{25}z_{39} - 4z_{39} + 4z_{39}^2$$

$$z_{41} = z_{10}^2 + 2z_{10}z_{11} + z_{11}^2$$

$$z_{42} = -2 - z_8^2 + 4z_8z_{41} - 2z_{10} + 2z_{10}^2 + 8z_{10}z_{41} + 2z_{25} - 4z_{41} + 4z_{41}^2$$

$$z_{43} = 31 + 16z_{10} + 16z_{10}z_{23} - 16z_{10}z_{25} + 8z_{23} - 12z_{23}^2 + 8z_{23}z_{25} - 24z_{25} + 4z_{25}^2 + 4z_{46}$$

$$z_{44} = 4z_{10} + 16z_{25}^2 + 32z_{25}z_{41} + 16z_{25}z_{42} + 16z_{25}z_{43} + 4z_{41} + 16z_{41}^2 + 16z_{41}z_{42} + 16z_{41}z_{43} + 2z_{42} + 4z_{42}^2 + 8z_{42}z_{43} + 4z_{43}^2$$

$$z_{45} = 16z_8^2 - 8z_8z_{10} - 32z_8z_{25} - 16z_8z_{28} - z_{10} + z_{10}^2 + 8z_{10}z_{25} + 4z_{10}z_{28} + 16z_{25}^2 + 16z_{25}z_{28} + 4z_{28}^2$$

$$z_{47} = 1 + 9z_{10}^2 - 24z_{10}z_{23} - 12z_{10}z_{46} + 16z_{23}^2 + 16z_{23}z_{46} + 4z_{46}^2$$

$$z_{49} = 1 + 4z_{48} + 4z_{48}^2$$

$$z_{50} = 1 + z_{10}^2 + 8z_{10}z_{48} + 4z_{10}z_{49} + 16z_{48}^2 + 16z_{48}z_{49} + 4z_{49}^2$$

$$z_{51} = -2 - z_9 + z_9^2 + 4z_9z_{10} + 4z_9z_{50} + 4z_{10}^2 + 8z_{10}z_{50} + 4z_{50}^2$$

$$z_{52} = 4z_{45}^2 + 8z_{45}z_{51} + 4z_{51}^2$$

$$z_{54} = 1 + 4z_{53} + 4z_{53}^2$$

$$z_{55} = z_{10}^2 + 8z_{10}z_{53} + 4z_{10}z_{54} + 16z_{53}^2 + 16z_{53}z_{54} + 4z_{54}^2$$

$$z_{56} = z_8 + z_8^2 + 4z_8z_{10} + 4z_8z_{55} + 4z_{10} + 4z_{10}^2 + 8z_{10}z_{55} + 4z_{55} + 4z_{55}^2$$

$$z_{57} = 10 + 12z_{56} + 4z_{56}^2$$

$$z_{58} = -2 - 4z_{10} - 16z_{10}z_{56} - 8z_{10}z_{57} + 4z_{23} + 16z_{23}z_{56} + 8z_{23}z_{57} + z_{46} - 8z_{56} + 16z_{56}^2 + 16z_{56}z_{57} - 4z_{57} + 4z_{57}^2$$

$$z_{59} = -z_9 + z_9^2 - 8z_9z_{10} + 8z_9z_{23} + 4z_9z_{58} + 16z_{10}^2 - 32z_{10}z_{23} - 16z_{10}z_{58} + 16z_{23}^2 + 16z_{23}z_{58} + 4z_{58}^2$$

$$z_{60} = 10 + 12z_{51} + 4z_{51}^2$$

$$z_{61} = -1 + 16z_{51}^2 + 16z_{51}z_{59} + 16z_{51}z_{60} + 4z_{59}^2 + 8z_{59}z_{60} + 4z_{60}^2$$

$$z_{76} = z_8z_9 - z_8z_{10} - z_9 - z_9z_{10} + z_{10} + z_{10}^2$$

$$z_{63} = 6z_8 + 3z_8^2 + 18z_8z_9 + 16z_9 + 27z_9^2 - 4z_9z_{10} - 3z_{10} - 3z_{10}^2 + 4z_{10}z_{19} + 19z_{19} + 8z_{19}^2 - z_{20} + 4z_{76}$$

$$z_{64} = -1 + 8z_8 + 4z_8z_{10} - 8z_8z_{63} + 16z_9 + 8z_9z_{10} - 16z_9z_{63} - z_{10} - z_{10}^2 - 4z_{10}z_{19} - 8z_{19} + 8z_{19}z_{63} - 3z_{63} + 4z_{63}^2$$

$$z_{65} = 2 + 12z_8 + 16z_8z_{10} - 16z_8z_{62} - 16z_8z_{64} + 24z_9 + 32z_9z_{10} - 32z_9z_{62} - 32z_9z_{64} - 4z_{10} - 4z_{10}^2 - 16z_{10}z_{19} - 12z_{19} + 16z_{19}z_{62} + 16z_{19}z_{64} + 4z_{62}^2 + 8z_{62}z_{64} + 4z_{63} + 2z_{64} + 4z_{64}^2$$

$$z_{66} = 2 + 4z_{65} + 4z_{65}^2$$

$$z_{67} = z_{13} + 2z_{13}z_{225}$$

$$z_{68} = z_{12} - z_{13} - 2z_{22}z_{67}$$

$$z_{69} = z_{12} - z_{13} - 4z_{24}z_{67} - 2z_{27}z_{67}$$

$$z_{75} = z_{12} - z_{13} - 2z_{44}z_{67} - 4z_{67}$$

$$z_{70} = -2z_{29}z_{67} + 2z_{44}z_{67} - 2z_{67} + z_{75}$$

$$z_{71} = z_{12} - z_{13} - 4z_{30}z_{67} - 2z_{32}z_{67}$$

$$z_{73} = z_{12} - z_{13} - 4z_{36}z_{67} - 2z_{37}z_{67}$$

$$z_{74} = z_{12} - z_{13} - 2z_{40}z_{67}$$

$$z_{77} = -1 - 2z_{11} - z_{11}^2 + z_{76}^2$$

$$z_{78} = z_{77}^2$$

$$z_{79} = 4z_{65}z_{67} + 2z_{66}z_{67} - 2z_{67}$$

$$z_{80} = z_{12}^2 - 2z_{12}z_{13} - 2z_{12}z_{79} + z_{13}^2 + 2z_{13}z_{79} + z_{79}^2$$

$$z_{81} = 1 + z_9 + 2z_9z_{225}$$

$$z_{82} = z_8 + 2z_8z_{225} - 2z_{225}$$

$$z_{83} = -1 - 2z_8 - 4z_8z_{225} - 4z_9 - 8z_9z_{225} + 2z_{19} + 4z_{19}z_{225} - 4z_{225}$$

$$z_{84} = z_{12}z_{82}$$

$$z_{85} = z_{13}z_{81} - z_{83}z_{84}$$

$$z_{86} = -z_{12}z_{81} - 6z_{13} + 8z_{13}z_{81} + 4z_{13}z_{82} + z_{13}z_{83} + 12z_{13}z_{225}$$

$$z_{87} = z_{12}z_{68} - z_{13}z_{68}$$

$$z_{88} = z_{69}z_{87}$$

$$z_{89} = z_{70}z_{88}$$

$$z_{90} = z_{71}z_{89}$$

$$z_{91} = z_{72}z_{90}$$

$$z_{92} = z_{73}z_{91}$$

$$z_{93} = z_{74}z_{92}$$

$$
\begin{aligned}
z_{94} &= z_{75} z_{93} \\
z_{95} &= z_{85} z_{94} \\
z_{96} &= z_{86} z_{95} \\
z_{97} &= z_{96}^2 \\
z_{98} &= z_2 + z_2 z_{16} - z_{16}
\end{aligned}
$$

$$
\begin{aligned}
z_{99} &= z_{14} z_{98} \\
z_{100} &= z_2 z_{15} - z_{15} + z_{15} z_{98} \\
z_{102} &= 1 - z_{140} z_{153} + z_{152} - 3 z_{153} + 6 z_{153} z_{220} \\
z_{127} &= z_{121} z_{213} \\
z_{130} &= z_{17} z_{127} \\
z_{155} &= -7 + z_{130} z_{226} - 2 z_{140} + 12 z_{220} \\
z_{103} &= 1 + z_{144} + 3 z_{145} + 6 z_{145} z_{220} - z_{146} - z_{155} z_{156} \\
z_{106} &= 3 z_{104} + 6 z_{104} z_{220}
\end{aligned}
$$

$$
\begin{aligned}
z_{105} &= z_{102} z_{211} - z_{106} z_{127} - z_{127} - 2 z_{127} z_{225} \\
z_{107} &= 4 + z_{140} - z_{140} z_{158} + z_{157} - 3 z_{158} + 6 z_{158} z_{220} - 6 z_{220} \\
z_{108} &= 3 z_{103} + 6 z_{103} z_{220} \\
z_{110} &= z_{107} z_{212} - z_{109} z_{130} \\
z_{113} &= z_{111} z_{212} - z_{112} z_{130} \\
z_{114} &= 1 - z_{140} z_{164} + z_{163} - 3 z_{164} + 6 z_{164} z_{220} \\
z_{116} &= z_{114} z_{212} - z_{115} z_{130} \\
z_{117} &= 1 - z_{140} z_{167} + z_{166} - 3 z_{167} + 6 z_{167} z_{220}
\end{aligned}
$$

$$
\begin{aligned}
z_{119} &= z_{117} z_{212} - z_{118} z_{130} \\
z_{120} &= 1 - z_{140} z_{170} + z_{169} - 3 z_{170} + 6 z_{170} z_{220} \\
z_{123} &= z_{120} z_{213} - z_{122} z_{130} \\
z_{126} &= z_{124} z_{213} - z_{125} z_{130} \\
z_{129} &= 1 - z_{128} z_{130} - z_{140} z_{180} + z_{179} - 3 z_{180} + 6 z_{180} z_{220} \\
z_{133} &= 1 - z_{140} z_{190} + z_{189} - 3 z_{190} + 6 z_{190} z_{220} \\
z_{135} &= -z_{130} z_{134} + z_{133} z_{211} \\
z_{138} &= -z_{130} z_{137} + z_{136} z_{212}
\end{aligned}
$$

$$
\begin{aligned}
z_{139} = {} & 36 + 6 z_0^2 + 12 z_0^2 z_{220} + 6 z_0 z_4 + 12 z_0 z_4 z_{220} + 3 z_1^2 + 6 z_1^2 z_{220} \\
& + 3 z_2^2 + 6 z_2^2 z_{220} + 3 z_4^2 + 6 z_4^2 z_{220} - 6 z_8 + 4 z_8^2 + 12 z_8^2 z_{220} \\
& + 2 z_8 z_9 + 12 z_8 z_9 z_{220} + 16 z_8 z_{19} + 48 z_8 z_{19} z_{220} + 8 z_8 z_{22} \\
& + 24 z_8 z_{22} z_{220} + 16 z_8 z_{24} + 48 z_8 z_{24} z_{220} + 8 z_8 z_{27} + 24 z_8 z_{27} z_{220} \\
& + 8 z_8 z_{29} + 24 z_8 z_{29} z_{220} + 16 z_8 z_{30} + 48 z_8 z_{30} z_{220} + 8 z_8 z_{32} \\
& + 24 z_8 z_{32} z_{220} + 8 z_8 z_{34} + 24 z_8 z_{34} z_{220} + 16 z_8 z_{36} + 48 z_8 z_{36} z_{220}
\end{aligned}
$$

$$
\begin{aligned}
&+ 8z_8z_{37} + 24z_8z_{37}z_{220} + 8z_8z_{40} + 24z_8z_{40}z_{220} + 8z_8z_{44} \\
&+ 24z_8z_{44}z_{220} + 16z_8z_{65} + 48z_8z_{65}z_{220} + 8z_8z_{66} + 24z_8z_{66}z_{220} \\
&- 12z_8z_{220} + 4z_9^2 + 12z_9^2z_{220} + 16z_9z_{19} + 48z_9z_{19}z_{220} + 8z_9z_{22} \\
&+ 24z_9z_{22}z_{220} + 16z_9z_{24} + 48z_9z_{24}z_{220} + 8z_9z_{27} + 24z_9z_{27}z_{220} \\
&+ 8z_9z_{29} + 24z_9z_{29}z_{220} + 16z_9z_{30} + 48z_9z_{30}z_{220} + 8z_9z_{32} \\
&+ 24z_9z_{32}z_{220} + 8z_9z_{34} + 24z_9z_{34}z_{220} + 16z_9z_{36} + 48z_9z_{36}z_{220} \\
&+ 8z_9z_{37} + 24z_9z_{37}z_{220} + 8z_9z_{40} + 24z_9z_{40}z_{220} + 8z_9z_{44} \\
&+ 24z_9z_{44}z_{220} + 16z_9z_{65} + 48z_9z_{65}z_{220} + 8z_9z_{66} + 24z_9z_{66}z_{220} \\
&+ 3z_{10}^2 + 6z_{10}^2z_{220} + 3z_{11}^2 + 6z_{11}^2z_{220} + 3z_{12}^2 + 6z_{12}^2z_{220} + 3z_{13}^2 \\
&+ 6z_{13}^2z_{220} + 3z_{14}^2 + 6z_{14}^2z_{220} + 3z_{15}^2 + 6z_{15}^2z_{220} + 3z_{16}^2 \\
&+ 6z_{16}^2z_{220} + 3z_{17}^2 + 6z_{17}^2z_{220} + z_{19} + 32z_{19}^2 + 96z_{19}^2z_{220} \\
&+ 32z_{19}z_{22} + 96z_{19}z_{22}z_{220} + 64z_{19}z_{24} + 192z_{19}z_{24}z_{220} + 32z_{19}z_{27} \\
&+ 96z_{19}z_{27}z_{220} + 32z_{19}z_{29} + 96z_{19}z_{29}z_{220} + 64z_{19}z_{30} + 192z_{19}z_{30}z_{220} \\
&+ 32z_{19}z_{32} + 96z_{19}z_{32}z_{220} + 32z_{19}z_{34} + 96z_{19}z_{34}z_{220} + 64z_{19}z_{36} \\
&+ 192z_{19}z_{36}z_{220} + 32z_{19}z_{37} + 96z_{19}z_{37}z_{220} + 32z_{19}z_{40} + 96z_{19}z_{40}z_{220} \\
&+ 32z_{19}z_{44} + 96z_{19}z_{44}z_{220} + 64z_{19}z_{65} + 192z_{19}z_{65}z_{220} + 32z_{19}z_{66} \\
&+ 96z_{19}z_{66}z_{220} + 8z_{22}^2 + 24z_{22}^2z_{220} + 32z_{22}z_{24} + 96z_{22}z_{24}z_{220} \\
&+ 16z_{22}z_{27} + 48z_{22}z_{27}z_{220} + 16z_{22}z_{29} + 48z_{22}z_{29}z_{220} + 32z_{22}z_{30} \\
&+ 96z_{22}z_{30}z_{220} + 16z_{22}z_{32} + 48z_{22}z_{32}z_{220} + 16z_{22}z_{34} + 48z_{22}z_{34}z_{220} \\
&+ 32z_{22}z_{36} + 96z_{22}z_{36}z_{220} + 16z_{22}z_{37} + 48z_{22}z_{37}z_{220} + 16z_{22}z_{40} \\
&+ 48z_{22}z_{40}z_{220} + 16z_{22}z_{44} + 48z_{22}z_{44}z_{220} + 32z_{22}z_{65} + 96z_{22}z_{65}z_{220} \\
&+ 16z_{22}z_{66} + 48z_{22}z_{66}z_{220} + 32z_{24}^2 + 96z_{24}^2z_{220} + 32z_{24}z_{27} \\
&+ 96z_{24}z_{27}z_{220} + 32z_{24}z_{29} + 96z_{24}z_{29}z_{220} + 64z_{24}z_{30} + 192z_{24}z_{30}z_{220} \\
&+ 32z_{24}z_{32} + 96z_{24}z_{32}z_{220} + 32z_{24}z_{34} + 96z_{24}z_{34}z_{220} + 64z_{24}z_{36} \\
&+ 192z_{24}z_{36}z_{220} + 32z_{24}z_{37} + 96z_{24}z_{37}z_{220} + 32z_{24}z_{40} + 96z_{24}z_{40}z_{220} \\
&+ 32z_{24}z_{44} + 96z_{24}z_{44}z_{220} + 64z_{24}z_{65} + 192z_{24}z_{65}z_{220} + 32z_{24}z_{66} \\
&+ 96z_{24}z_{66}z_{220} + 8z_{27}^2 + 24z_{27}^2z_{220} + 16z_{27}z_{29} + 48z_{27}z_{29}z_{220} \\
&+ 32z_{27}z_{30} + 96z_{27}z_{30}z_{220} + 16z_{27}z_{32} + 48z_{27}z_{32}z_{220} + 16z_{27}z_{34} \\
&+ 48z_{27}z_{34}z_{220} + 32z_{27}z_{36} + 96z_{27}z_{36}z_{220} + 16z_{27}z_{37} + 48z_{27}z_{37}z_{220} \\
&+ 16z_{27}z_{40} + 48z_{27}z_{40}z_{220} + 16z_{27}z_{44} + 48z_{27}z_{44}z_{220} + 32z_{27}z_{65} \\
&+ 96z_{27}z_{65}z_{220} + 16z_{27}z_{66} + 48z_{27}z_{66}z_{220} + 8z_{29}^2 + 24z_{29}^2z_{220} \\
&+ 32z_{29}z_{30} + 96z_{29}z_{30}z_{220} + 16z_{29}z_{32} + 48z_{29}z_{32}z_{220} + 16z_{29}z_{34} \\
&+ 48z_{29}z_{34}z_{220} + 32z_{29}z_{36} + 96z_{29}z_{36}z_{220} + 16z_{29}z_{37} + 48z_{29}z_{37}z_{220} \\
&+ 16z_{29}z_{40} + 48z_{29}z_{40}z_{220} + 16z_{29}z_{44} + 48z_{29}z_{44}z_{220} + 32z_{29}z_{65} \\
&+ 96z_{29}z_{65}z_{220} + 16z_{29}z_{66} + 48z_{29}z_{66}z_{220} + 32z_{30}^2 + 96z_{30}^2z_{220} \\
&+ 32z_{30}z_{32} + 96z_{30}z_{32}z_{220} + 32z_{30}z_{34} + 96z_{30}z_{34}z_{220} + 64z_{30}z_{36} \\
&+ 192z_{30}z_{36}z_{220} + 32z_{30}z_{37} + 96z_{30}z_{37}z_{220} + 32z_{30}z_{40} + 96z_{30}z_{40}z_{220} \\
&+ 32z_{30}z_{44} + 96z_{30}z_{44}z_{220} + 64z_{30}z_{65} + 192z_{30}z_{65}z_{220} + 32z_{30}z_{66} \\
&+ 96z_{30}z_{66}z_{220} + 8z_{32}^2 + 24z_{32}^2z_{220} + 16z_{32}z_{34} + 48z_{32}z_{34}z_{220} \\
&+ 32z_{32}z_{36} + 96z_{32}z_{36}z_{220} + 16z_{32}z_{37} + 48z_{32}z_{37}z_{220} + 16z_{32}z_{40} \\
&+ 48z_{32}z_{40}z_{220} + 16z_{32}z_{44} + 48z_{32}z_{44}z_{220} + 32z_{32}z_{65} + 96z_{32}z_{65}z_{220} \\
&+ 16z_{32}z_{66} + 48z_{32}z_{66}z_{220} + 8z_{34}^2 + 24z_{34}^2z_{220} + 32z_{34}z_{36} \\
&+ 96z_{34}z_{36}z_{220} + 16z_{34}z_{37} + 48z_{34}z_{37}z_{220} + 16z_{34}z_{40} + 48z_{34}z_{40}z_{220}
\end{aligned}
$$

$$
\begin{aligned}
&+ 16z_{34}z_{44} + 48z_{34}z_{44}z_{220} + 32z_{34}z_{65} + 96z_{34}z_{65}z_{220} + 16z_{34}z_{66}\\
&+ 48z_{34}z_{66}z_{220} + 32z_{36}^2 + 96z_{36}^2z_{220} + 32z_{36}z_{37} + 96z_{36}z_{37}z_{220}\\
&+ 32z_{36}z_{40} + 96z_{36}z_{40}z_{220} + 32z_{36}z_{44} + 96z_{36}z_{44}z_{220} + 64z_{36}z_{65}\\
&+ 192z_{36}z_{65}z_{220} + 32z_{36}z_{66} + 96z_{36}z_{66}z_{220} + 8z_{37}^2 + 24z_{37}^2z_{220}\\
&+ 16z_{37}z_{40} + 48z_{37}z_{40}z_{220} + 16z_{37}z_{44} + 48z_{37}z_{44}z_{220} + 32z_{37}z_{65}\\
&+ 96z_{37}z_{65}z_{220} + 16z_{37}z_{66} + 48z_{37}z_{66}z_{220} + 8z_{40}^2 + 24z_{40}^2z_{220}\\
&+ 16z_{40}z_{44} + 48z_{40}z_{44}z_{220} + 32z_{40}z_{65} + 96z_{40}z_{65}z_{220} + 16z_{40}z_{66}\\
&+ 48z_{40}z_{66}z_{220} + 8z_{44}^2 + 24z_{44}^2z_{220} + 32z_{44}z_{65} + 96z_{44}z_{65}z_{220}\\
&+ 16z_{44}z_{66} + 48z_{44}z_{66}z_{220} + 32z_{65}^2 + 96z_{65}^2z_{220} + 32z_{65}z_{66}\\
&+ 96z_{65}z_{66}z_{220} + 8z_{66}^2 + 24z_{66}^2z_{220} + 3z_{103}^2 + 6z_{103}^2z_{220} + 3z_{121}^2\\
&+ 6z_{121}^2z_{220} + 3z_{127}^2 + 6z_{127}^2z_{220} - z_{145}^2 - z_{153}^2 + 2z_{158} - z_{158}^2\\
&- z_{161}^2 - z_{164}^2 - z_{167}^2 - z_{170}^2 - z_{173}^2 - z_{177}^2 - z_{180}^2 - z_{183}^2 - z_{187}^2\\
&- z_{190}^2 + z_{193}^2 + 3z_{195}^2 + 6z_{195}^2z_{220} - z_{214}^2 - z_{215}^2 - z_{216} - z_{216}^2\\
&+ 186z_{220} + 324z_{220}^2 + 216z_{220}^3 + 24z_{220}z_{225} + 24z_{220}z_{225}^2 + 12z_{225} + 12z_{225}^2
\end{aligned}
$$

$$z_{142} = -z_{139} + 2z_{139}z_{141}$$

$$z_{143} = z_{139}z_{142} + z_{142}$$

$$z_{154} = z_{140}z_{153} + 7z_{153}$$

$$z_{159} = -7 - z_{140} + z_{140}z_{158} + 7z_{158}$$

$$z_{160} = -1 + z_{111} + z_{140}z_{161} + 3z_{161} - 6z_{161}z_{220}$$

$$z_{162} = z_{140}z_{161} + 7z_{161}$$

$$z_{165} = z_{140}z_{164} + 7z_{164}$$

$$z_{168} = z_{140}z_{167} + 7z_{167}$$

$$z_{171} = z_{140}z_{170} + 7z_{170}$$

$$z_{174} = z_{140}z_{173} + 7z_{173}$$

$$z_{172} = -1 + z_{121} + z_{155}z_{175} - 3z_{173} - 6z_{173}z_{220} + z_{174}$$

$$z_{176} = -1 + z_{124} + z_{140}z_{177} + 3z_{177} - 6z_{177}z_{220}$$

$$z_{178} = z_{140}z_{177} + 7z_{177}$$

$$z_{181} = z_{140}z_{180} + 7z_{180}$$

$$z_{186} = -1 + z_{130}z_{131} + z_{132} + z_{140}z_{187} + 3z_{187} - 6z_{187}z_{220}$$

$$z_{188} = z_{140}z_{187} + 7z_{187}$$

$$z_{191} = z_{140}z_{190} + 7z_{190}$$

$$z_{192} = -1 + z_{136} + z_{140}z_{193} + 3z_{193} - 6z_{193}z_{220}$$

$$z_{194} = z_{140}z_{193} + 7z_{193}$$

$$z_{199} = 1 + z_{192} + z_{197}z_{205}$$

$$
\begin{aligned}
z_{200} = {} & z_8 + z_9 + 4z_{19} + 2z_{22} + 4z_{24} + 2z_{27} + 2z_{29} + 4z_{30} + 2z_{32} + 2z_{34} + 4z_{36} + 2z_{37} + 2z_{40} + 2z_{44}\\
& + 4z_{65} + 2z_{66} + 4z_{193}z_{206}
\end{aligned}
$$

$$z_{207} = z_{140}z_{198} + 7z_{198}$$

$$z_{208} = z_{200} + z_{200}z_{201}$$

Retained residuals

$$R_0 = -1 - z_2 z_5 + z_6 z_7 - 2z_{225}$$

$$R_1 = 1 + z_9 - z_9^2 - 4z_9 z_{10} - 4z_9 z_{47} - 4z_{10}^2 - 8z_{10} z_{47} - 4z_{47}^2 + z_{48}$$

$$R_2 = -9 + 7z_8 - z_8^2 - 8z_8 z_{45} - 4z_8 z_{52} + 24z_{45} - 16z_{45}^2 - 16z_{45} z_{52} + 12z_{52} - 4z_{52}^2 + z_{53}$$

$$R_3 = z_{10} - z_{10}^2 - 8z_{10} z_{59} - 4z_{10} z_{61} - 16z_{59}^2 - 16z_{59} z_{61} - 4z_{61}^2 + z_{62}$$

$$R_4 = z_{12} - z_{13} - 2z_{34} z_{67} + 6z_{67} - z_{72}$$

$$R_5 = -2z_1^2 + 2z_1 z_{12} + 2z_1 z_{13} + 2z_1 z_{99} + 2z_1 z_{100} - z_{12}^2 - 2z_{12} z_{99} - z_{13}^2 - 2z_{13} z_{100} + z_{17} z_{101} - z_{78} z_{97} - z_{80} z_{97} - z_{99}^2 - z_{100}^2$$

$$R_6 = -1 + z_{142}^2 - z_{143}^2 + z_{223}^2$$

$$R_7 = 2z_{144} + z_{144}^2 + z_{145}^2 - z_{146}^2$$

$$R_8 = 2z_{152} + z_{152}^2 + z_{153}^2 - z_{154}^2$$

$$R_9 = -39 + z_{130} z_{226} - 8z_{140} - 12z_{140} z_{220} - 36z_{220} + 36z_{220}^2$$

$$R_{10} = 1 + 2z_{157} + z_{157}^2 - 2z_{158} + z_{158}^2 - z_{159}^2$$

$$R_{11} = 2z_{160} + z_{160}^2 + z_{161}^2 - z_{162}^2$$

$$R_{12} = 2z_{163} + z_{163}^2 + z_{164}^2 - z_{165}^2$$

$$R_{13} = 2z_{166} + z_{166}^2 + z_{167}^2 - z_{168}^2$$

$$R_{14} = 2z_{169} + z_{169}^2 + z_{170}^2 - z_{171}^2$$

$$R_{15} = 2z_{172} + z_{172}^2 + z_{173}^2 - z_{174}^2$$

$$R_{16} = 2z_{176} + z_{176}^2 + z_{177}^2 - z_{178}^2$$

$$R_{17} = 2z_{179} + z_{179}^2 + z_{180}^2 - z_{181}^2$$

$$R_{18} = -1 + 2z_{182} + z_{182}^2 + z_{183}^2 - 2z_{184} - z_{184}^2$$

$$R_{19} = z_{127} + z_{155}z_{185} - z_{182} - 3z_{183} - 6z_{183}z_{220} + z_{184}$$

$$R_{20} = 2z_{186} + z_{186}^2 + z_{187}^2 - z_{188}^2$$

$$R_{21} = 2z_{189} + z_{189}^2 + z_{190}^2 - z_{191}^2$$

$$R_{22} = 2z_{192} + z_{192}^2 + z_{193}^2 - z_{194}^2$$

$$R_{23} = -1 + z_{197}^2 + z_{198}^2 - z_{207}^2$$

$$R_{24} = -1 + z_{199}^2 + z_{200}^2 - z_{208}^2$$

$$\begin{aligned} R_{25} = {} & -2 + 2z_0^2 + 2z_0z_4 - 2z_0z_{145} - 2z_0z_{153} + z_1^2 - 2z_1z_{183} + z_2^2 - 2z_2z_{173} \\ & + z_4^2 - 2z_4z_{153} + z_8^2 - 2z_8z_{158} - 2z_8z_{193} + z_9^2 - 2z_9z_{161} - 2z_9z_{193} \\ & + z_{10}^2 - 2z_{10}z_{164} + z_{11}^2 - 2z_{11}z_{167} + z_{12}^2 - 2z_{12}z_{170} + z_{13}^2 - 2z_{13}z_{177} \\ & + z_{14}^2 - 2z_{14}z_{180} + z_{15}^2 - 2z_{15}z_{187} + z_{16}^2 - 2z_{16}z_{190} - 8z_{19}z_{193} \\ & - 4z_{22}z_{193} - 8z_{24}z_{193} - 4z_{27}z_{193} - 4z_{29}z_{193} - 8z_{30}z_{193} - 4z_{32}z_{193} \\ & - 4z_{34}z_{193} - 8z_{36}z_{193} - 4z_{37}z_{193} - 4z_{40}z_{193} - 4z_{44}z_{193} - 8z_{65}z_{193} \\ & - 4z_{66}z_{193} - z_{140}z_{210} + z_{144}^2 + z_{152}^2 + z_{157}^2 + 2z_{158} + z_{160}^2 + z_{163}^2 \\ & + z_{166}^2 + z_{169}^2 + z_{172}^2 + z_{176}^2 + z_{179}^2 + z_{182}^2 + z_{186}^2 + z_{189}^2 + z_{192}^2 \\ & + 2z_{193}^2 - 6z_{210} - z_{214}^2 - z_{215}^2 - z_{216} - z_{216}^2 \end{aligned}$$

$$R_{26} = -z_{103}z_{211} + z_{127}$$

$$R_{27} = -z_{108}z_{212} + z_{127}$$

$$
\begin{aligned}
R_{28} = {} & -1 - 8z_8z_{19} - 4z_8z_{22} - 8z_8z_{24} - 4z_8z_{27} - 4z_8z_{29} - 8z_8z_{30} - 4z_8z_{32} \\
& - 4z_8z_{34} - 8z_8z_{36} - 4z_8z_{37} - 4z_8z_{40} - 4z_8z_{44} - 8z_8z_{65} - 4z_8z_{66} \\
& - 8z_9z_{19} - 4z_9z_{22} - 8z_9z_{24} - 4z_9z_{27} - 4z_9z_{29} - 8z_9z_{30} - 4z_9z_{32} \\
& - 4z_9z_{34} - 8z_9z_{36} - 4z_9z_{37} - 4z_9z_{40} - 4z_9z_{44} - 8z_9z_{65} - 4z_9z_{66} - z_{19} \\
& - 16z_{19}^2 - 16z_{19}z_{22} - 32z_{19}z_{24} - 16z_{19}z_{27} - 16z_{19}z_{29} - 32z_{19}z_{30} \\
& - 16z_{19}z_{32} - 16z_{19}z_{34} - 32z_{19}z_{36} - 16z_{19}z_{37} - 16z_{19}z_{40} - 16z_{19}z_{44} \\
& - 32z_{19}z_{65} - 16z_{19}z_{66} - 4z_{22}^2 - 16z_{22}z_{24} - 8z_{22}z_{27} - 8z_{22}z_{29} - 16z_{22}z_{30} \\
& - 8z_{22}z_{32} - 8z_{22}z_{34} - 16z_{22}z_{36} - 8z_{22}z_{37} - 8z_{22}z_{40} - 8z_{22}z_{44} - 16z_{22}z_{65} \\
& - 8z_{22}z_{66} - 16z_{24}^2 - 16z_{24}z_{27} - 16z_{24}z_{29} - 32z_{24}z_{30} - 16z_{24}z_{32} - 16z_{24}z_{34} \\
& - 32z_{24}z_{36} - 16z_{24}z_{37} - 16z_{24}z_{40} - 16z_{24}z_{44} - 32z_{24}z_{65} - 16z_{24}z_{66} - 4z_{27}^2 \\
& - 8z_{27}z_{29} - 16z_{27}z_{30} - 8z_{27}z_{32} - 8z_{27}z_{34} - 16z_{27}z_{36} - 8z_{27}z_{37} - 8z_{27}z_{40} \\
& - 8z_{27}z_{44} - 16z_{27}z_{65} - 8z_{27}z_{66} - 4z_{29}^2 - 16z_{29}z_{30} - 8z_{29}z_{32} - 8z_{29}z_{34} \\
& - 16z_{29}z_{36} - 8z_{29}z_{37} - 8z_{29}z_{40} - 8z_{29}z_{44} - 16z_{29}z_{65} - 8z_{29}z_{66} - 16z_{30}^2 \\
& - 16z_{30}z_{32} - 16z_{30}z_{34} - 32z_{30}z_{36} - 16z_{30}z_{37} - 16z_{30}z_{40} - 16z_{30}z_{44} \\
& - 32z_{30}z_{65} - 16z_{30}z_{66} - 4z_{32}^2 - 8z_{32}z_{34} - 16z_{32}z_{36} - 8z_{32}z_{37} - 8z_{32}z_{40} \\
& - 8z_{32}z_{44} - 16z_{32}z_{65} - 8z_{32}z_{66} - 4z_{34}^2 - 16z_{34}z_{36} - 8z_{34}z_{37} - 8z_{34}z_{40} \\
& - 8z_{34}z_{44} - 16z_{34}z_{65} - 8z_{34}z_{66} - 16z_{36}^2 - 16z_{36}z_{37} - 16z_{36}z_{40} - 16z_{36}z_{44} \\
& - 32z_{36}z_{65} - 16z_{36}z_{66} - 4z_{37}^2 - 8z_{37}z_{40} - 8z_{37}z_{44} - 16z_{37}z_{65} - 8z_{37}z_{66} \\
& - 4z_{40}^2 - 8z_{40}z_{44} - 16z_{40}z_{65} - 8z_{40}z_{66} - 4z_{44}^2 - 16z_{44}z_{65} - 8z_{44}z_{66} \\
& - 16z_{65}^2 - 16z_{65}z_{66} - 4z_{66}^2 - z_{145}^2 - z_{153}^2 + 2z_{158} - z_{158}^2 - z_{161}^2 \\
& - z_{164}^2 - z_{167}^2 - z_{170}^2 - z_{173}^2 - z_{177}^2 - z_{180}^2 - z_{183}^2 - z_{187}^2 - z_{190}^2 \\
& + z_{193}^2 - z_{214}^2 - z_{215}^2 - z_{216} - z_{216}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{29} = {} & -1 - 4z_{105}^2 - 4z_{110}^2 - 4z_{113}^2 - 4z_{116}^2 - 4z_{119}^2 - 4z_{123}^2 - 4z_{126}^2 + z_{127}^2 \\
& - 4z_{129}^2 - 4z_{132}^2 - 4z_{135}^2 - 4z_{138}^2 - z_{217}^2 - z_{218}^2 - z_{219} - z_{219}^2
\end{aligned}
$$

$$
R_{30} = 5 + z_{140} + 8z_{141} - 8z_{141}^2 - z_{223}^2
$$

The required equation is therefore exactly

$$
D_{83}(x_0, \dots, x_{82}) = \sum_{j=0}^{30} R_j^2 = 0.
$$

The electronic source preserves the source-row indices, unit-pivot witnesses and binary graph, so these printed definitions can be checked independently against the literal coefficient files.

References

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